VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Belagavi-590014, Karnataka

A Synopsis

on

" Advanced AI-Based Phishing website Detection System with Real-time Analysis "

A dissertation work submitted towards

in partial fulfillment of the requirement for the award of the degree of

Bachelor of Engineering

in

Information Science and Engineering

Submitted by

Aishwarya (3GN23IS007)

Mayuri (3GN23IS023)

Under the guidance of

Prof. Komal

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

GURU NANAK DEV ENGINEERING COLLEGE BIDAR

MAILOOR ROAD, BIDAR-585403, KARNATAKA

Academic Year: 2025-26

Phone: 08482- 226949 **Website: www.gndecb.ac.in**

GURU NANAK DEV ENGINEERING COLLEGE

BIDAR-585403, KARNATAKA

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Report entitled "**Phishing Website Detector**" is a bonafide work carried out **Aishwarya (3GN23IS007), Mayuri (3GN23IS023)** in practical fulfillment of the requirements for the award of Bachelor of Engineering in **VI Semester in B.E in Information Science and Engineering of Visvesvaraya Technological University, Belagavi** during the year 2025-26. It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the synopsis report deposited in the department library. The synopsis report has been approved as it satisfies the academic requirements in respect of major project work prescribed for the said degree.

Signature of Guide
Prof. komal

Signature of HOD
Dr. Masarath B

ACKNOWLEDGEMENT

The major project report on **"Phishing Website Detector"** is the outcome of guidance, moral support and devotion bestowed onus throughout our work. Fort this we acknowledge and express out profound sense of gratitude and thanks to everybody who have been a source of inspiration during the project work

First and foremost, we offer our sincere phrases of thanks with innate humility to our Principal **Dr.Veena S**, who has been a constant source of support and encouragement. We feel deeply indebted to our H.O.D **Dr.Masarath B** for right help provided from the time of inception till date. I would take this opportunity to acknowledge our **Prof. Komal** who not only stood by us as a source of inspiration, but also dedicated his time to enable us to present the project on time. We would be failing in Endeavour, if we do not thank our Co-ordinator, **Prof. Sangameshwar K** who have helped us in every aspect of our work.

1. **Mayuri (3GN23IS023)**
2. **Aishwarya (3GN23IS007)**

Project Synopsis

Abstract

Introduction

Objective

Scope of Project

Existing System

Disadvantages

Proposed System

Advantages

Project Outcome

System Requirements:

Hardware Requirements:

Software Requirements:

Conclusion

References

Signature along with date

Guide	**Project Coordinator**	**HOD ISE**

Title: Advanced AI-Based Phishing Website Detection System with Real-time Analysis.

Broad Area: Artificial Intelligence and Machine learning

Project Synopsis

Abstract

The surge in internet usage has fueled a dramatic rise in cybercrime, with phishing attacks being a major threat. These attacks use fraudulent websites mimicking legitimate entities to steal sensitive user data (passwords, credit cards). Traditional rule-based or blacklist detection methods are now inadequate against sophisticated phishing.

This project proposes an **Advanced AI-Based Phishing Website Detection System** using Machine Learning for real-time, high-accuracy identification. Unlike conventional detectors, it employs a multi-layered feature analysis:

1. **URL-Based Analysis:** Examines structural features like length, special characters, subdomains, and IP address usage.
2. **Domain-Based Analysis:** Evaluates domain age, WHOIS data, DNS integrity, SSL certificate validity, and reputation scores.
3. **Content-Based Analysis (NLP):** Uses Natural Language Processing to analyze webpage content for suspicious keywords, form fields, and brand mismatches.

Introduction

1.1 Background

The internet has become an indispensable part of modern civilization — from online banking and e-commerce to healthcare records and government services. This

widespread digital dependence has simultaneously created a fertile ground for cybercriminals. Among all forms of cybercrime, phishing has consistently ranked as the most common and damaging attack. The FBI's IC3 reports phishing as the #1 reported cybercrime, resulting in billions of dollars in annual losses.

Phishing exploits not just technical vulnerabilities but human psychology (social engineering). Even technically aware users fall victim to well-crafted pages that visually replicate trusted platforms with pixel-perfect precision.

1.2 What is Phishing?

Phishing is a cyberattack where a malicious actor creates a fraudulent website impersonating a trusted entity — a bank, e-commerce site, or government portal — to steal sensitive data such as:

- Login credentials
- Credit/debit card numbers
- PII (Aadhaar, PAN, DOB)
- OTPs and security codes

Types of Phishing:

Type	Description
URL/Website Phishing	Fake websites mimicking real ones — this project's focus
Email Phishing	Deceptive emails with malicious links
Spear Phishing	Targeted attacks on specific individuals
Whaling	Highly targeted spear phishing aimed at senior executives (C-level)

Type	Description
Smishing / Vishing	Phishing conducted via SMS (text message) or voice calls
Pharming	DNS manipulation or host file compromise to redirect users to malicious sites, even when typing the correct URL
Clone Phishing	Creating an exact replica of a legitimate, previously sent email with updated, malicious links

Objectives

2.1 Overview

Objectives are structured across three levels:

Primary — core goals that define project success
Secondary — enhancement goals that add depth
Measurable — quantifiable benchmarks to evaluate completion

2.2 Primary Objectives

Objective 1 — Study & Analyze Phishing Characteristics

Conduct in-depth study of distinguishing characteristics of phishing websites across URL structure, domain properties, and webpage content.

Understanding what makes a URL phishing-like is the foundation — without this, no ML algorithm can compensate.

Objective 2 — Build a Multi-Layer Feature Extraction Engine

Tier	Features Extracted
URL-Based	URL length, presence of '@' symbol, count of subdomains, use of IP address instead of domain name, URL entropy, suspicious TLDs
Domain-Based	Domain age, WHOIS registration data integrity, DNS record analysis, SSL certificate validity, external blacklist status/reputation scores
Content/NLP	Phishing keywords and phrases, presence of login forms, ratio of internal to external links, title/brand mismatch, suspicious anchor text patterns

Scope of the Project

3.1 What is Scope?

Scope defines the exact boundaries of the project — what it will do, what it won't do, who it serves, and what technology and data it covers.

3.2 Functional Scope

#	Feature
1	URL submission via web interface
2	URL feature extraction (length, subdomains, special chars, entropy)
3	Domain feature extraction (WHOIS, DNS, SSL certificate validity)

#	Feature
4	NLP-based webpage content analysis (keywords, forms, link ratio)
5	ML-based binary classification (Phishing / Legitimate)

3.3 User Scope

User Type	Usage
General Internet Users	Verify suspicious links from emails/WhatsApp
Students & Researchers	Study ML models, phishing patterns
Security Enthusiasts	Analyze URL feature reasoning
Developers	Base for browser extension / API

Existing System

Existing phishing detection systems fall into 6 major categories:

- Blacklist-Based Detection
- Heuristic / Rule-Based Detection
- Visual Similarity-Based Detection
- Early / Limited ML-Based Detection
- Browser Built-in Protection
- Third-Party Security APIs

Disadvantages

1. **Completely Reactive:** Blacklists only block already-discovered URLs hours to days after launch, leaving a window during which thousands of users fall victim. Zero-day phishing is invisible.
2. **Easily Bypassed:** Minor URL changes (subdomains, encoding, parameters, shorteners) create infinite unblocked variants because blacklists store exact URLs.

3. **Short-Lived Domains Retire Undetected:** Fast-flux phishing domains are registered, used, and decommissioned within 12-24 hours, completing the damage before blocklist detection can occur.
4. **Massive Maintenance Overhead:** Manual verification by large security teams creates a time-lagged bottleneck, ensuring the blacklist is always behind active campaigns.

Proposed System

The proposed system is an Advanced AI-Based Phishing Website Detection System that uses AI, ML, and NLP to identify phishing websites in real-time. It moves beyond a single URL check by unifying analysis across three essential tiers: the **URL's structure**, the **domain's properties**, and the **webpage's content**. System Architecture — Five Stages

1. **URL Input and Validation:** Accepts the URL, validates its format and reachability, and flags invalid inputs.
2. **Three-Tier Feature Extraction:** Extracts a minimum of 20 features simultaneously across the **URL, Domain, and Content** tiers.
3. **Machine Learning Classification:** Feeds the feature vector into a trained ML model (expected to be **XGBoost**), which assigns a binary label (Phishing/Legitimate) and a **Risk Score (0-100)**.

Advantages

1. **Proactive Detection:** Uses machine learning to find new phishing URLs, unlike reactive blacklist-based methods.
2. **Multi-Layer Analysis:** Harder for attackers to bypass by analyzing URL structure, domain, and content together.
3. **High Accuracy:** Machine learning achieves >95% accuracy, outperforming rule-based tools.
4. **Interpretable Results:** Provides feature-level explanations and a Risk Score (0-100) for clarity on flagging.
5. **Low False Positives:** Holistic feature evaluation reduces incorrect blocking of

legitimate sites common in heuristic tools.

Project Outcome

1. **High-Accuracy ML Model:** A trained ML model (XGBoost/Random Forest/Logistic Regression) classifying URLs as Phishing or Legitimate with $\geq$95% accuracy.
2. **Automated Feature Extraction:** A pipeline extracting URL, domain, and NLP features automatically from any submitted URL.
3. **Real-Time Risk Scoring:** A 0-100 Risk Score providing immediate threat level understanding for every URL.
4. **Comparative Model Report:** Detailed performance comparison (Accuracy, F1, Precision, Recall, AUC-ROC) of Logistic Regression, Random Forest, and XGBoost models.
5. **Transparent Explanation Report:** A clear report for each analysis, listing suspicious features and explaining the phishing classification in plain language.

System Requirements

Hardware:

1. **Processor:** Intel Core i3 minimum (i5/i7 recommended for faster training).
2. **RAM:** Minimum 4 GB (8 GB+ recommended for smooth training/scraping).
3. **Storage:** Minimum 5 GB free disk space for data, models, and project files.
4. **Internet:** Required for fetching live URL data (WHOIS, DNS, SSL) during feature extraction.

Software:

1. **OS:** Windows 10/11, Ubuntu, or macOS (must support Python 3.x).
2. **Programming Language:** Python 3.8 or above.
3. **ML Libraries:** Scikit-learn (Logistic Regression, Random Forest), XGBoost (gradient boosting).
4. **Data Handling:** Pandas and NumPy for dataset management and feature

engineering.

5. **NLP/Web Scraping:** BeautifulSoup4, Requests for crawling; NLTK for NLP/keyword analysis.
6. **Web Framework:** Flask for the local web application interface.
7. **Visualization:** Matplotlib and Seaborn for plotting performance charts (AUC-ROC, confusion matrices).
8. **Browser:** Modern browser (Chrome, Firefox, Edge) to access the Flask application.

References

A. Sahingoz, E. Batur, F. Bulut, and A. K. Ilhan, "Machine learning based phishing detection from URLs," Expert Systems with Applications, vol. 117, pp. 345-357, 2019. (Reference for URL-based feature extraction).

S. Abu-Nimeh, D. Nappa, X. Wang, and S. Nair, "A comparison of machine learning techniques for phishing detection," in Proceedings of the anti-phishing working groups 2nd annual eCrime researchers summit, 2007, pp. 60-69. (Reference for comparing ML models like Logistic Regression and Random Forest).

R. Shirazi, K. N. N. Tun, and C. H. Hsu, "XGBoost for cyber security anomaly detection," IEEE Access, vol. 8, pp. 112345-112355, 2020. (Reference for the effectiveness of the XGBoost model).

M. N. Sakib and M. M. Rahman, "Phishing Website Detection Using Natural Language Processing and Machine Learning," International Journal of Computer Applications, vol. 182, no. 48, pp. 15-20, 2019. (Reference for using NLP on webpage content).

Y. A. Alsirhani, M. M. Alshammari, and A. S. Alshammari, "Real-time phishing detection utilizing domain name system and SSL certificate features," IEEE/ACM Transactions on Networking, 2021. (Reference for Domain and SSL analysis).

Conclusion

In conclusion, the proposed Advanced AI-Based Phishing Website Detection System offers a comprehensive and proactive approach to cybersecurity. By moving beyond traditional, static URL blacklisting and adopting a multi-layered feature extraction methodology—combining URL heuristics, domain registry details, and NLP-driven content analysis—the system is uniquely equipped to identify sophisticated, zero-day phishing attacks. The comparative analysis of machine learning algorithms, particularly XGBoost and Random Forest, ensures optimal real-time detection accuracy. Ultimately, through its user-friendly Flask-based web interface, this project not only delivers a robust technical solution for threat classification but also provides an accessible tool to protect everyday internet users from increasingly deceptive online fraud. As a scalable architecture, it also lays a strong foundation for future integration into browser extensions for seamless, automated protection.